

Nishant Acharya

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EDUCATION

University of California, Davis

Davis, CA

Doctor of Philosophy in Computer Science. Anticipated Jun. 2028

University of Massachusetts

Amherst, MA

Bachelor of Science in Computer Science. May 2021

PROJECTS

Reverse IP Geolocation | *Active Measurement, Data Engineering*

Jun. 2024 – Present

- Ran active measurements using 25 RIPE Atlas probes per library across 1,071 libraries, covering approximately 3.2 million /24 net ranges filtered from 29.4 million candidate prefixes
- Reconciled partially aligned data sources (FCC broadband, Census, WHOIS), applying statistical filtering to reduce false positives and improve network property inference precision

Quantifying the Impact of IXP Shutdowns | *Network Verification, Traffic Engineering*

Sep. 2024 – Present

- Designed a passive measurement framework to analyze BGP control plane disruptions at backbone interconnection points and quantify downstream impact on global traffic distribution and path quality
- Identified novel approaches to infer unknown links in IXP-adjacent networks, improving topology verification accuracy
- Extended framework with topology-aware quantification using metAScritic; collaborated with cross-institutional team of 3; presented at IMC 2025 Student Workshop and PIMF 2025

EXPERIENCE

ISOC Pulse Fellow

Jun. 2025 – Dec. 2025

Internet Society

Davis, CA

- Developed a BGP control plane analysis framework to quantify routing centrality and traffic path distribution at IXPs
- Integrated topology modeling to identify unmapped active AS links near IXbr São Paulo, improving accuracy of large-scale network topology inference
- Applied statistical validation to assess topology inference accuracy; presented at IMC 2025 and PIMF 2025

Graduate Student Researcher

Sep. 2021 – Present

Internet Measurement and Policy Lab, UC Davis

Davis, CA

- Designed and deployed scalable measurement frameworks for large-scale data collection, BGP routing analysis, and network anomaly detection across backbone Internet infrastructure
- Collaborated with research teams across 3 institutions to develop and refine experimental methodologies, producing open-source frameworks and conference publications
- Mentored 2 undergraduate students through a summer research program, supporting their first empirical research contributions

Teaching Assistant, Courses 152 / 150 / 32B

Winter 2022 – Winter 2026

University of California, Davis

Davis, CA

- Supported instruction of Computer Networks, Data Structures, and Operating Systems for 100+ students per year
- Built autograders spanning server-client communication to TCP Reno implementation on a noisy channel, used by 150 to 170 students in C++ and Python

PUBLICATIONS AND SUBMISSIONS

[1] **Nishant Acharya**, Vasileios Giotsas, Amreesh Phokeer, and Alexander Gamero-Garrido. *Quantifying the impact of IXP peering disruptions*. IMC 2025 Student Workshop; PIMF Spain 2025

[2] **Nishant Acharya**, Anyu Yang, Ashutosh Kshirsagar, Humaira Fasih Ahmed Hashmi, Jiayi Liu, Elizabeth Belding, David Choffnes, Alexander Gamero-Garrido. *Reverse IP Geolocation*. Preparing for Resubmission

[3] **Nishant Acharya**, Vasileios Giotsas, Amreesh Phokeer, and Alexander Gamero-Garrido. *[Title withheld — under review]*. Submitted at HotNets 2026

SKILLS

Languages: Python, C/C++, Java, Bash

Tools: RIPE Atlas, Git, Docker, pandas, NumPy, SciPy, Linux

Networking: BGP, IP Routing, DNS, OSI Model, Traceroutes, Network Topology Analysis, Traffic Path Analysis

Research Methods: Large-Scale Data Collection, Statistical Analysis, Experimental Design, Computational Modeling